

styles? Unfortunately, no. This kind of toolchain is good for one-page documents. Or, multi-page documents without headers and footers. If you want those, then you have to use a different tool called wkhtmltopdf. This is essentially a headless Firefox browser that can convert heavily formatted HTML documents to PDF. You can specify headers and footers using separate HTML files. I created all 29 of my books using this tool. It supports lots of CSS styles that LibreOffice does not support. My books would not look so rich if I relied on LibreOffice. (wkhtmltopdf is based on an old Firefox codebase that is not being updated. A lone Indian is maintaining it. It has a few bugs. Read my Unicode tricks article on how I fixed them.) Also, unlike LibreOffice, wkhtmltopdf can execute JavaScript. It has an option for adding a few seconds of delay so that the JavaScript can do its thing before wkhtmltopdf starts printing the document to PDF. So, for simple documents, use LibreOffice. And, for heavy-duty documents, use wkhtmltopdf. Both these programs will use the headings in the source document to create the bookmark tree in the PDF document. LibreOffice has one advantage that wkhtmltopdf does not have — it can create ePUB ebooks and several other types of documents.

Additional document features

If you have to merge two ODT or DOCX documents, I do not know of or care for any tool to do it. Do the merging in the markdown source material and then create the combined document.

For PDFs, there are lots of tools. For my books, I have to combine some page-size images with the PDF of the interior pages. For that, I use ImageMagick and pdftk.

```
magick title-page.png -resize 100% front.pdf
pdftk front.pdf jokes.pdf output book.pdf
```

pdftk is a very powerful tool and does more than merging PDF pages or collating them from different documents. It can also watermark and encrypt your PDF.

```
pdftk book.pdf output book-encrypted.pdf \
    encrypt_128bit \
    owner_pw RChRdStlMn^012 \
    user_pw FrSfTWrfnDtn^321
```

And, for a touch of class, add metadata to the PDF:

```
echo "InfoBegin" > meta.txt
echo "InfoKey: Title" >> meta.txt
echo "InfoValue: 2020 Jokebook by V. Subhash" >> meta.txt
```

```
echo "InfoBegin" >> meta.txt
echo "InfoKey: Subject" >> meta.txt
echo "InfoValue: Fresh Clean Jokes" >> meta.txt
```

```
echo "InfoBegin" >> meta.txt
echo "InfoKey: Author" >> meta.txt
echo "InfoValue: V. Subhash (&#169; 2022 V. Subhash. All
rights reserved.)" >> meta.txt
```

```
pdftk "jokebook.pdf" update_info meta.txt output
2020-jokebook.pdf
```

Whatever I cannot do with these free PDF tools, I go ahead and write my own custom utility using iText Java. For example, when I found that pdftk was unable to handle some of my bigger books, I used a JAR file executable that I had created using iText.

Use in online applications

Document creation requires a lot of heavy lifting in terms of CPU usage, disk access and memory requirements. Do not integrate these tools directly with web applications. Use a daemon (background process or service) that is launched by the OS on startup to do the document creation. Your web applications should just queue document-creation jobs to this daemon. When the documents are ready, the daemon should invoke an API routine in the web application to notify that the job has been completed. Otherwise, your online users will crash your system when their numbers build up.

Donate

If free software such as those mentioned in this article helped you to reduce costs, then make it a point to donate some money to their projects. If you are an independent software vendor (ISV), then tell your client that you did not write all the code and some part of the system relies on free tools. Tell the client to make donations commensurate to the usage of the free software in that system. The creator of iText has an incredible story (check the *Reference* given at the end of this article) of how Google refused to offer nothing more than a T-shirt and a mug for using his PDF library in their products such as Google Analytics, Google Docs and Google Calendar. Do not be like that. Indian corporations have a statutory obligation to divert a certain amount of their profits to charitable causes. Most of them will be glad to route donations to open source projects as CSR (corporate social responsibility) grants. **END** 🐧

Reference

- <https://entrepreneurd.lowagie.com/chapters/c13.html>

By: V. Subhash

The author has written over two dozen books including Quick Start Guide To FFmpeg (from Apress), Linux Command-Line Tips & Tricks, CommonMark Ready Reference, Cool Electronic Projects, How To Install Solar and Unlikely Stories.